Seth Grable, PhD

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Research Profile

Theoretical physicist (PhD, CU Boulder, 2026) specializing in nonperturbative QCD and Yang–Mills theory, with first-author publications in JHEP and Physics Letters B. Develops analytic and semi-analytic methods for strongly interacting field theories, including confinement in many-flavor QCD and the Yang–Mills vacuum. Particularly interested in singular learning theory, where degenerate saddles, zero modes, and phase transitions, central to nonperturbative field theory, govern how neural networks learn. Seeking to apply these methods to machine learning and AI research.

Education

PhD, Physics — University of Colorado Boulder 2022 – 2026
Nonperturbative methods for non-Abelian gauge theories and QCD. GPA 3.90

PhD studies, Physics — Colorado School of Mines 2021 – 2022
Thermodynamics and transport in open quantum systems. GPA 4.00

MSt, Philosophy of Physics — University of Oxford 2019 – 2020
Focus on quantum foundations

BA, Physics, Magna Cum Laude — Case Western Reserve University 2016 – 2019
GPA 3.87

Research Experience

Nonperturbative QCD and Yang–Mills Theory — CU Boulder 2021 – 2026
Advisor: Prof. Paul Romatschke

- Developed analytic and semi-analytic solutions to strongly interacting quantum field theories in regimes where perturbation theory breaks down.
- Built and maintained Python and Mathematica codes for large numerical calculations; validated results against known limits and independent numerical evaluations.
- Published as first author in *JHEP* and *Physics Letters B*; further manuscript under review at *Nuclear Physics B*.

Transport in Open Quantum Systems — Colorado School of Mines 2021 – 2022
Advisor: Prof. Lincoln Carr

- Modeled energy and particle transport through quantum spin chains coupled to reservoirs using Lindblad master equations, the standard framework for decoherence and noise in qubits.
- Investigated an out-of-equilibrium quantum analogue of the Onsager reciprocal relations.

Quantum Algorithms and Quantum Machine Learning — Case Western Reserve University 2020 – 2021
Advisor: Prof. Shuai Xu

- Developed classical algorithms inspired by HHL (quantum linear-system solver) with a computer science group working on quantum circuits, algorithms, and machine learning.

***PT*-Symmetric Quantum Mechanics** — Case Western Reserve University 2018 – 2020
Advisor: Prof. Harsh Mathur

- Studied a *PT*-symmetric form of the contact interaction and its applications to condensed-matter many-body systems.

Technical Skills

- **Quantum & physics:** open quantum systems, quantum field theory, statistical mechanics and transport, quantum foundations, electromagnetism.
- **Quantitative:** linear algebra, differential equations, probability and statistics, optimization; building and validating models of complex systems.
- **Computing:** Python (NumPy, pandas, PyTorch), Mathematica, LaTeX.
- **Communication:** peer-reviewed writing, conference talks, technical explanation for non-specialists and leader-

ship.

Publications

- S. Grable, "Nonperturbative Analytic Methods for Quantum Field Theories" Diss. University of Colorado at Boulder, 2026
- S. Grable and P. Romatschke, "Vanishing Polyakov loop for QCD with twelve massless quarks," *Phys. Lett. B* **862**, 139310 (2025). arXiv:2310.12203
- S. Grable, "An almost complete Yang–Mills calculation," *The XVIth Quark Confinement and the Hadron Spectrum Conference*, 040 (2025).
- S. Grable and M. Weiner, "A fully solvable model of fermionic interaction in $3+1d$," *Journal of High Energy Physics* 2023.9 (2023): 17. arXiv:2302.08603
- S. Grable, "Interacting CFTs for all couplings: thermal versus entanglement entropy at large N," *JHEP* **10**, 133 (2022). arXiv:2205.15383

Preprints

- S. Grable and J. Barcelona, "Scalar charges in a self-dual background," (Under review JHEP) arXiv:2607.07938 (2026).
- S. Grable, "Ghost hunting in the Yang–Mills vacuum," (under review NUC PHYS B) arXiv:2607.07954 (2026).
- S. Grable, "An analytic Yang–Mills vacuum calculation in 3+1d," arXiv:2407.13042 (2024).

Invited and Contributed Talks

- *Yang–Mills Beyond One-Loop Order* — invited talk, QCD Vacuum Structure and Confinement, 2024.
- *An Almost Complete Yang–Mills Calculation* — XVIth Quark Confinement and the Hadron Spectrum, 2024.
- *Semiclassical Solution of Twelve-Flavor Chiral QCD in the Continuum* — Cracow School of Theoretical Physics, 2023.

Teaching and Mentoring

Lecturer — University of Colorado Boulder 2022 – 2026

- Electricity & Magnetism (PHYS 1120), Physics of Energy & the Environment (PHYS 3070), Newtonian Mechanics; full course responsibility for lectures, exams, and grading.
- Thermal QFT lecture series for students and faculty, Colorado School of Mines (2021–2022).
- Mentored three undergraduate researchers; one project led to a co-authored publication.

Awards

Case Alumni Junior/Senior Scholarship, CWRU (2017–2019) · Betty Gardner Physics Scholarship, University of Akron (2015–2016)